Visual-Inertial SLAM

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I. Introduction

In the field of autonomous robots, two fundamental questions are essential for performing any meaningful task: "Where am I?" and "What is around me?" These two questions, more formally, give rise to the tasks of localization and mapping, respectively.

The Simultaneous Localization and Mapping (SLAM) problem concerns itself with a robot's ability to answer both of those questions at once. In this project, we address the problem of Visual-Inertial SLAM using data from *Clearpath Jackal Robots* navigating the MIT campus in Cambridge. The goal of this project is to use the IMU and camera sensors on this robot to enhance the trajectory estimation of the robot.

We do this using the EKF. First, we do the *prediction step* of the kalman filter to achieve localization. Then, we estimate landmark positions surrounding the robot using the *update step* of the kalman filter to achieve mapping. Finally, these two steps are combined to perform SLAM.

II. Problem Formulation

We model a robot that has an IMU and a stereo camera moving in a three-dimensional environment. Let $\{W\}$ denote the world frame and $\{I\}$ denote the IMU frame. In this project, we take the IMU frame to be the body frame. The pose of the IMU at time t relative to the world frame is represented by a pose matrix:

$$T_t := {}_W T_{I,t} = \begin{pmatrix} R_t & p_t \\ 0^\top & 1 \end{pmatrix} \in SE(3), \tag{1}$$

where $R_t \in SO(3)$ is the orientation of the IMU and $p_t \in \mathbb{R}^3$ is its position in the world frame.

The IMU provides measurements of linear velocity $v_t \in \mathbb{R}^3$ and angular velocity $\omega_t \in \mathbb{R}^3$ both being expressed in the IMU frame. We define the input to be

$$u_t = \begin{bmatrix} v_t \\ \omega_t \end{bmatrix} \in \mathbb{R}^6. \tag{2}$$

The robot motion is modeled probabilistically as

$$T_{t+1} = f(T_t, u_t, w_t), \tag{3}$$

where $f(\cdot)$ denotes the kinematics and $w_t \sim \mathcal{N}(0, W)$ is a zero-mean Gaussian process noise.

Under the kinematic model used here, we have:

$$f(T_t, u_t, w_t) = T_t \exp(\tau_t \hat{u}_t + \hat{w}_t)\,, \tag{4}$$

where τ_t is the sampling interval and $\hat{u}_t, \hat{w}_t \in \mathfrak{se}(3)$ are the matrix representations of the twist input and process noise, respectively.

Let $m_j \in \mathbb{R}^3$ denote the world frame coordinates of the j-th landmark, for $j = 1, \ldots, M$. We use a homogeneous representation

$$\bar{m}_j := \begin{bmatrix} m_j \\ 1 \end{bmatrix} \in \mathbb{R}^4. \tag{5}$$

At time t, the stereo camera provides image measurements of visible landmarks. For landmark j, the corresponding landmark observation is

$$z_{t,i} = \begin{bmatrix} u_{L,t,i} \\ v_{L,t,i} \\ u_{R,t,i} \\ v_{R,t,i} \end{bmatrix} \in \mathbb{R}^4, \tag{6}$$

where $(u_{L,t,i}, v_{L,t,i})$ and $(u_{R,t,i}, v_{R,t,i})$ are the pixel coordinates of the matched feature in the left and right images, respectively.

Let K_s be the stereo calibration matrix and let ${}_O T_I \in SE(3)$ denote the deduced extrinsic transformation from the IMU frame to the camera optical frame. The observation model is given probabilistically by [1]

$$z_{t,i} = h(T_t, \bar{m}_j, v_{t,i}), \tag{7}$$

where

$$h(T_t, \bar{m}_j, v_{t,i}) = K_s \pi\big({}_O T_I T_t^{-1} \bar{m}_j\big) + v_{t,i}, \tag{8}$$

and the measurement noise is assumed Gaussian,

$$v_{t,i} \sim \mathcal{N}(0, V). \tag{9}$$

Here, $\pi(\cdot)$ denotes the perspective projection in homogeneous coordinates:

$$\pi(q) = \frac{1}{q_3} q, \qquad q \in \mathbb{R}^4. \tag{10}$$

Not all landmarks are observed at every timestep due to occlusions or limited camera field of view and for such landmarks we have $z_{t,i} = [-1, -1, -1, -1]^T$. We assume that the data association between landmarks and image measurements is known.

The objective of the visual-inertial SLAM problem is to estimate the robot trajectory

$$\{T_t\}_{t=0}^{T} \tag{11}$$

and the landmark positions

$$\{m_j\}_{j=1}^{M} \tag{12}$$

from the sequence of collected IMU measurements $\{u_t\}_{t=0}^{T-1}$ and stereo feature observations $\{z_{t,i}\}$.

Because of our Gaussian motion and measurement noise assumptions, this problem can be stated as the minimization of a joint SLAM objective [2]:

$$\min_{\{T_t\},\{m_j\}} \sum_{t=1}^{K} \|z_t - h(T_t, m)\|_2^2 + \sum_{t=0}^{K-1} \|T_{t+1} - f(T_t, u_t)\|_2^2 . \tag{13}$$

The first term penalizes the difference between predicted and observed stereo feature measurements, while the second term penalizes deviations from the IMU-based motion model.

III. Technical Approach

The Visual-Inertial SLAM problem is solved using and extended kalman filter (EKF). We would like to estimate both the trajectory and the landmark locations simultaneously by fusing inertial measurements with visual feature observations. We first solve each problem individually; then, we combine to reach a true SLAM approach.

The pipeline consists of three stages:

- IMU localization via EKF prediction
- Landmark mapping via EKF update
- Visual–Inertial SLAM

In each stage, graphics are produced to illustrate the results. These plots allow us to visualize the accuracy of the estimations.

Throughout this section, the pose of the IMU with respect to the world frame is represented as

$$T_t = {}_W T_{I,t} = \begin{bmatrix} R_t & p_t \\ 0^T & 1 \end{bmatrix} \in SE(3)$$

where $R_t \in SO(3)$ is the orientation and $p_t \in \mathbb{R}^3$ represents the position of the IMU in the world frame – as previously mentioned.

A. Gaussian Distribution on $SE(3)$

Because the pose lies on the $SE(3)$ manifold, we represent uncertainty using a perturbation in the Lie Algebra. A random pose can be written as [3]

$$T = \mu \exp(\hat{\delta\mu})$$

where

$$\delta\mu \sim \mathcal{N}(0, \Sigma).$$

The EKF estimates

$$\mu_{t|t} \in SE(3), \qquad \Sigma_{t|t} \in \mathbb{R}^{6\times 6}.$$

B. IMU Localization via EKF Prediction

In this first section, the robot trajectory is estimated using only IMU measurements. The IMU provides linear velocity v_t and angular velocity ω_t, which form

$$u_t = \begin{bmatrix} v_t \\ \omega_t \end{bmatrix} \in \mathbb{R}^6.$$

The corresponding Lie algebra representation is

$$\hat{u}_t = \begin{bmatrix} \hat{\omega}_t & v_t \\ 0^T & 0 \end{bmatrix}.$$

The continuous-time kinematics of a rigid body are

$$\dot{T}(t) = T(t)\hat{u}(t).$$

Assuming constant velocity over the interval τ_t, the discrete-time motion model becomes

$$T_{t+1} = T_t \exp(\tau_t \hat{u}_t).$$

This equation integrates the measured velocities to propagate the robot pose forward in time.

1) Prediction Step: Given the prior distribution

$$T_t | z_{0:t}, u_{0:t-1} \sim \mathcal{N}(\mu_{t|t}, \Sigma_{t|t}),$$

the predicted mean pose is

$$\mu_{t+1|t} = \mu_{t|t} \exp(\tau_t \hat{u}_t).$$

This step propagates the pose estimate using the IMU measurements.

Because the IMU measurements are noisy, uncertainty must also be propagated. The perturbation dynamics are

$$\delta\mu_{t+1|t} = \exp(-\tau_t u_t^{\wedge})\delta\mu_{t|t} + \sqrt{\tau_t} w_t$$

where

$$u_t^{\wedge} = \begin{bmatrix} \hat{\omega}_t & \hat{v}_t \\ 0 & \hat{\omega}_t \end{bmatrix}$$

and

$$w_t \sim \mathcal{N}(0, W).$$

Taking expectations yields the covariance prediction

$$\Sigma_{t+1|t} = \exp(-\tau_t u_t^{\wedge})\Sigma_{t|t} \exp(-\tau_t u_t^{\wedge})^T + \tau_t W.$$

Then, a 2D plot of the estimated trajectory is generated. The trajectory is extracted from the translation components of the pose matrices and plotted in the (x, y) plane. This choice is natural because the robot is not experiencing any vertical movement. This visualization allows us to observe the path estimated using only the IMU motion model.

C. Landmark Mapping via EKF Update

In the second stage, we estimate the positions of the landmarks observed by the stereo camera. During this stage, the robot poses are treated as known and are obtained via the previous step.

Let

$$m_j \in \mathbb{R}^3$$

denote the world-frame coordinates of landmark j. We assume that the landmarks are static.

The stereo camera provides pixel observations

$$z_{t,i} = \begin{bmatrix} u_L \\ v_L \\ u_R \\ v_R \end{bmatrix} \in \mathbb{R}^4.$$

1) Observation Model: A landmark is first transformed into the camera frame

$$q = {}_O T_I T_t^{-1} m_j$$

where ${}_O T_I$ is the transformation from the IMU frame to the camera optical frame. We assume that the data association function $\Delta_t : \{1, \ldots, M\} \to \{1, \ldots, N_t\}$ is known, meaning each observed feature $z_{t,i}$ corresponds to a specific landmark m_j. This assumption allows the EKF update to directly incorporate feature measurements, leading to the following stereo projection model

$$z_{t,i} = h(T_t, m_j) + v_{t,i}$$

with

$$h(T_t, m_j) = K_s \pi(q)$$

where K_s is the stereo calibration matrix and $\pi(\cdot)$ is the perspective projection function.

Stacking all observations gives

$$z_t = K_s \pi({}_O T_I T_t^{-1} m) + v_t.$$

2) EKF Update Step: Given the prior landmark estimate

$$m | z_{0:t} \sim \mathcal{N}(\mu_t, \Sigma_t),$$

the predicted observation is

$$\tilde{z}_{t+1,i} = K_s \pi({}_O T_I T_{t+1}^{-1} \mu_{t,j}).$$

The innovation

$$z_{t+1} - \tilde{z}_{t+1}$$

represents the difference between the predicted and observed feature locations.

The observation model is linearized using the Jacobian

$$H_{t+1,i,j} = K_s \frac{\partial \pi}{\partial q} ({}_O T_I T_{t+1}^{-1} \mu_{t,j}) {}_O T_I T_{t+1}^{-1} P^T$$

where

$$P = [I \quad 0].$$

The Kalman gain is

$$K_{t+1} = \Sigma_t H_{t+1}^T (H_{t+1} \Sigma_t H_{t+1}^T + I \otimes V)^{-1}.$$

The landmark estimates are updated as

$$\mu_{t+1} = \mu_t + K_{t+1}(z_{t+1} - \tilde{z}_{t+1})$$

$$\Sigma_{t+1} = (I - K_{t+1} H_{t+1}) \Sigma_t.$$

This step refines the landmark positions so that the predicted projections align with the observed stereo features.

The results of the this stage are visualized by plotting the landmarks together with the IMU trajectory. All the landmark measurements are used. Each landmark is displayed as a point in the world frame.

D. Visual–Inertial SLAM

The last stage estimates both the trajectory and the landmark positions simultaneously via an EKF prediction and update step. Instead of assuming the IMU trajectory is correct, visual observations are used to refine the robot pose and the surrounding map.

The joint state is

$$x_t = \begin{bmatrix} T_t \\ m \end{bmatrix}.$$

The covariance matrix includes cross-correlations between robot pose uncertainty and landmark uncertainty

$$\Sigma_t = \begin{bmatrix} \Sigma_{rr} & \Sigma_{rl} \\ \Sigma_{lr} & \Sigma_{ll} \end{bmatrix}.$$

1) Prediction: During prediction only the robot pose evolves according to the IMU motion model

$$\mu_{t+1|t,r} = \mu_{t|t,r} \exp(\tau_t \hat{u}_t).$$

The covariance propagation updates both the robot covariance and the cross-covariances as follows

$$\Sigma_{t+1|t} = \begin{bmatrix} F_t \Sigma_{rr} F_t^T + W & F_t \Sigma_{rl} \\ \Sigma_{lr} F_t^T & \Sigma_{ll} \end{bmatrix}.$$

2) Update: When visual observations are available, both the robot pose and the landmarks are updated.

The predicted observation is

$$\tilde{z}_{t+1,i} = K_s \pi({}_O T_I \mu_{t+1|t,r}^{-1} \mu_{t,j,l}).$$

The Kalman gain is

$$K_{t+1} = \Sigma_{t+1|t} H_{t+1}^T (H_{t+1} \Sigma_{t+1|t} H_{t+1}^T + I \otimes V)^{-1}.$$

The landmark estimates are updated linearly

$$\mu_{t+1|t+1,l} = \mu_{t,l} + K_{t+1}(z_{t+1} - \tilde{z}_{t+1})$$

while the robot pose is updated using

$$\mu_{t+1|t+1,r} = \mu_{t+1|t,r} \exp((K_{t+1}(z_{t+1} - \tilde{z}_{t+1}))^{\wedge}).$$

where the $\exp(\cdot)$ function denotes the matrix exponential that maps elements of the Lie algebra $\mathfrak{se}(3)$ to the Lie group $SE(3)$.

This joint update allows visual observations to correct drift in the inertial motion model while simultaneously improving the landmark map.

The results are visualized by plotting the trajectories and map from the VI-SLAM trajectory together with the IMU-only trajectory and the estimated landmarks.

IV. Results

In this section, the results of each stage are shared for the two datasets provided with the assignment. For each stage we visualize the estimated trajectory and landmark map to evaluate the performance of the algorithm. Note: in each stage we use:

	Value
Angular Noise	0.01
Linear Noise	0.01
Pixel Noise	8.0

TABLE I: Noise Values for EKF

A. IMU Localization via EKF Prediction

In this stage, the EKF prediction step propagates the robot pose using the measured linear and angular velocities. Because no visual measurements are incorporated at this stage, the trajectory is completely determined by integrating the IMU motion model.

Figures 1 and 2 show the estimated trajectories using IMU localization for Dataset 00 and Dataset 01. The trajectories are extracted from the translation component of the pose matrices and plotted in the (x, y) plane.

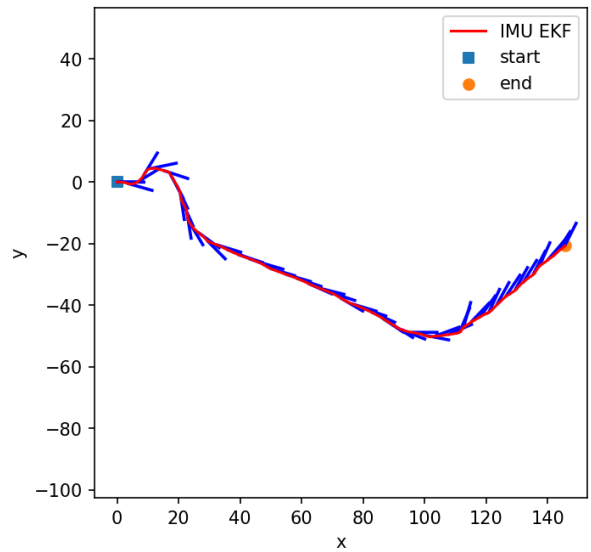

Fig. 1: Estimated IMU trajectory using EKF prediction for Dataset 00.

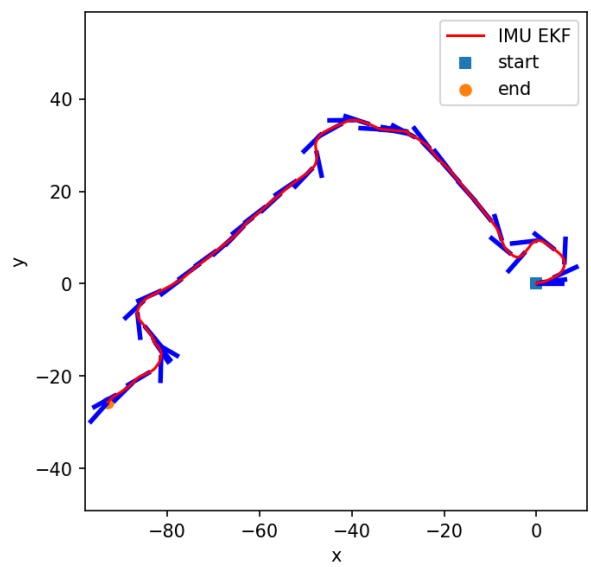

Fig. 2: Estimated IMU trajectory using EKF prediction for Dataset 01.

The trajectories seem reasonable given that there are no extremely jagged movements depicted. However, because inertial measurements accumulate noise, the trajectory may be drifting from the true path. This drift motivates the use of visual observations to correct the estimate in section IV-C

B. Landmark Mapping via EKF Update

In the second stage we estimate the positions of landmarks observed by the stereo camera. The robot poses obtained from section IV-A are treated as known, and the EKF update step is used to refine the landmark positions based on visual observations.

Figures 3 and 4 show the estimated landmark positions together with the IMU trajectory for the two datasets.

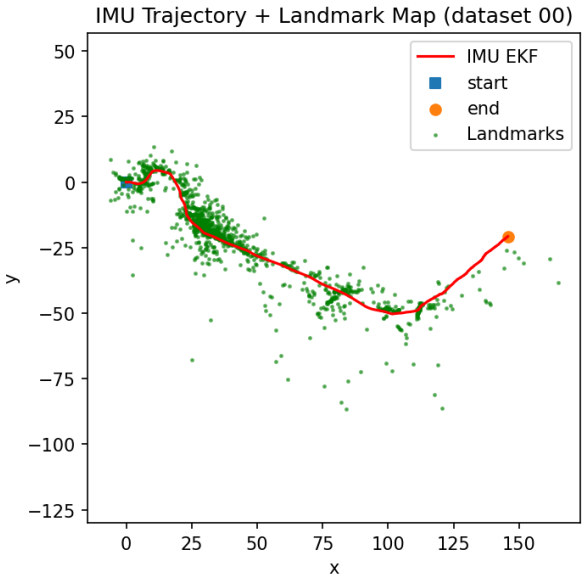

Fig. 3: Estimated landmark map and IMU trajectory for Dataset 00.

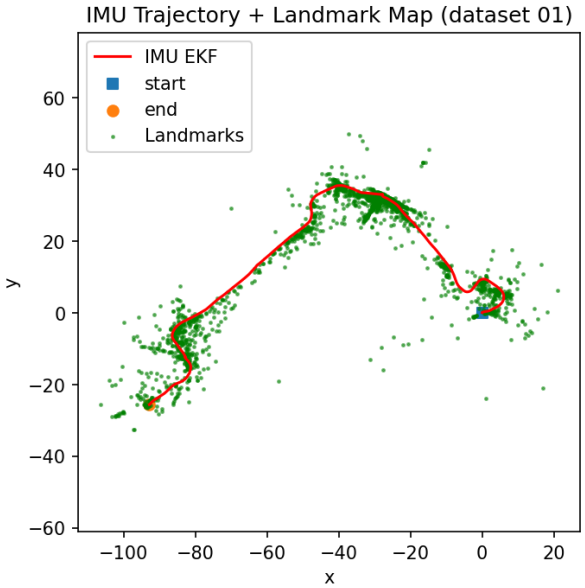

Fig. 4: Estimated landmark map and IMU trajectory for Dataset 01.

The resulting landmark distributions reflect the spatial structure of the environment observed by the camera. Because the robot trajectory is treated as fixed during this stage, any error in the IMU localization may affect the accuracy of the landmark positions. For both datasets, the landmark mapping seems feasible in the sense that they are mostly clustered around the trajectory.

C. Visual–Inertial SLAM

The final stage jointly estimates both the robot trajectory and the landmark positions. In contrast to the previous stages, visual observations are used to update both the robot pose and the map. This allows visual measurements to correct drift in the IMU motion model.

Figures 5 and 6 show the final Visual–Inertial SLAM results for the two datasets. The plots illustrate the estimated robot trajectory along with the landmark positions obtained from the joint EKF estimation.

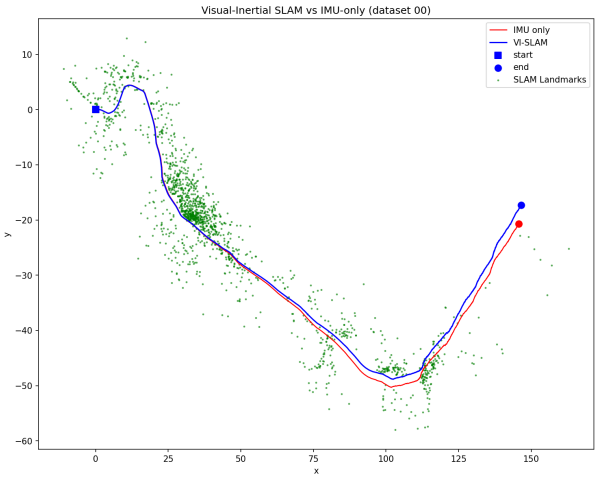

Fig. 5: Visual–Inertial SLAM results for Dataset 00.

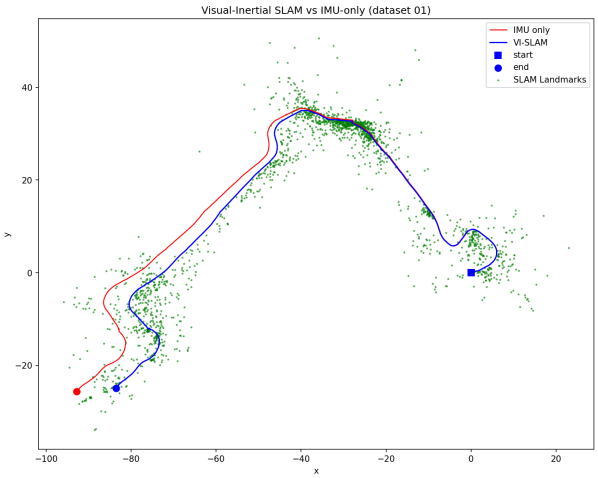

Fig. 6: Visual–Inertial SLAM results for Dataset 01.

Compared to the IMU-only trajectory (red), the Visual-Intertial SLAM results demonstrate improved consistency due to the incorporation of visual elements. The blue line seems to try to center itself between large clusters of landmarks, which physically makes sense: in most situations, there are usually landmarks on both sides of a vehicle. Overall, the SLAM trajectory hugs the IMU-only trajectory closely, while making small improvements along the way.

V. Conclusion

In this project, the Visual-Inertial SLAM problem was solved through and extended kalman filter framework. The approach combines IMU measurements with visual observations in order to estimate the trajectory and position of landmarks in an environment.

The results demonstrate that SLAM produces trajectories that better align with the structure of the observed environment and reduces the drift present in the IMU-only estimates.
[1]

References

[1] N. Atanasov, "Ece276a: Sensing and estimation in robotics – lecture 12: Visual-inertial slam." https://natanaso.github.io/ece276a/ref/ECE276A_12_VI_SLAM.pdf, 2024. University of California, San Diego.

[2] N. Atanasov, "Ece276a: Sensing and estimation in robotics – lecture 5: Factor graph slam." Lecture slides, University of California San Diego, 2024. Accessed: 2026.

[3] N. Atanasov, "Ece276a: Sensing and estimation in robotics – matrix lie groups." Lecture slides, University of California San Diego, 2024. Accessed: 2026.

[1]ChatGPT (OpenAI) was used as a clarification tool during this project as well as for help with LaTeX formatting issues.